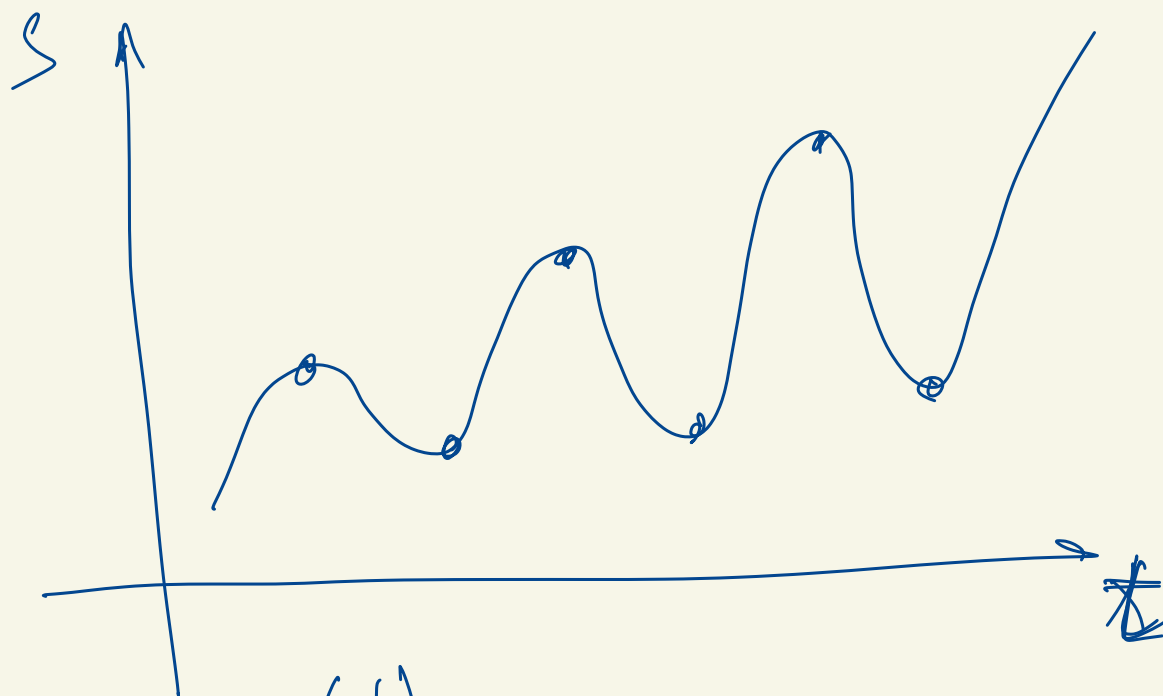
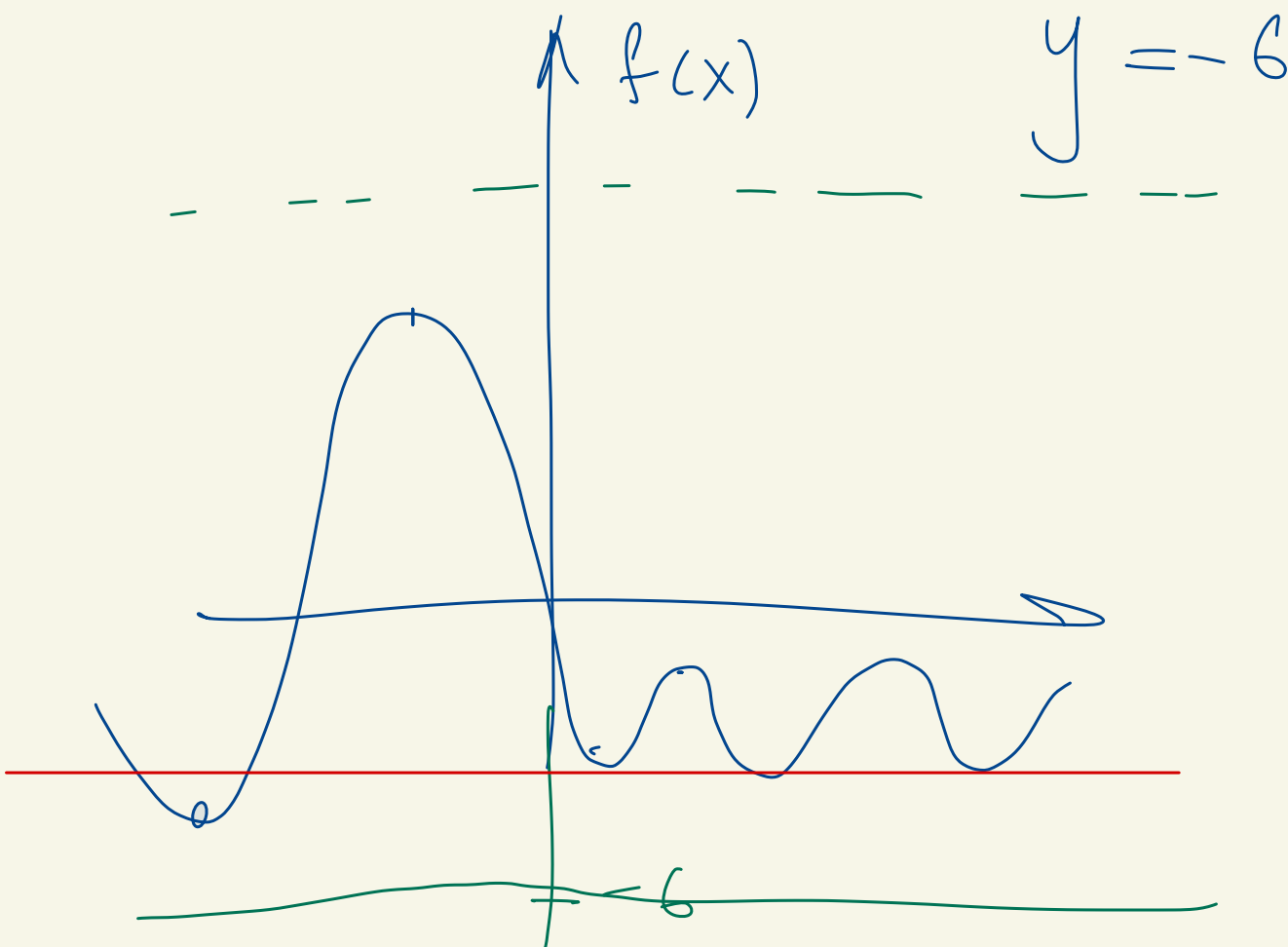


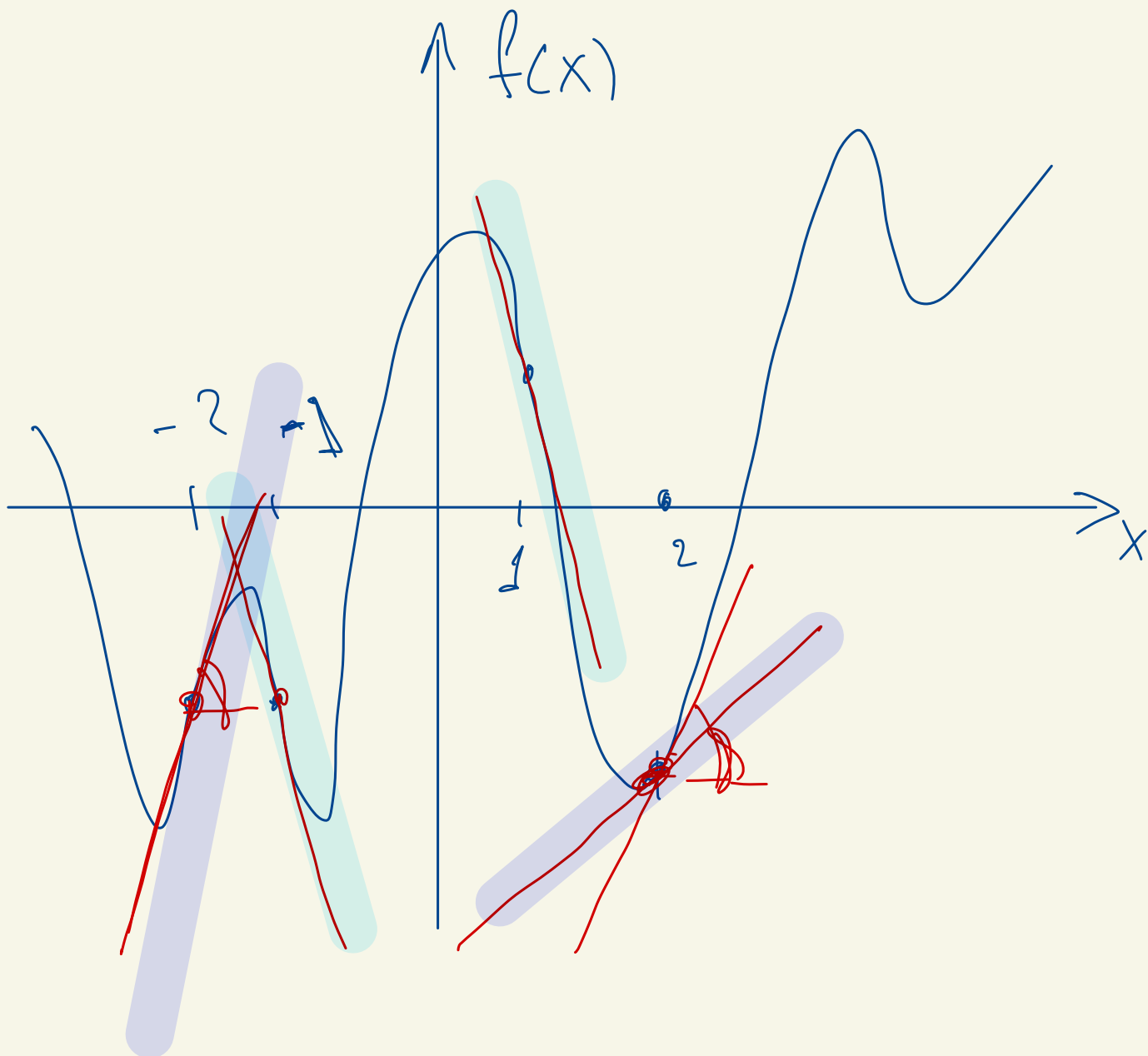




$$S(t) = x(t)$$

$$\dot{S}(t) = v(t) = 0 \Rightarrow \exists \text{ экстремум}$$





$$|x^2 - 2x - 3| - 2a = |x - a| - 1$$

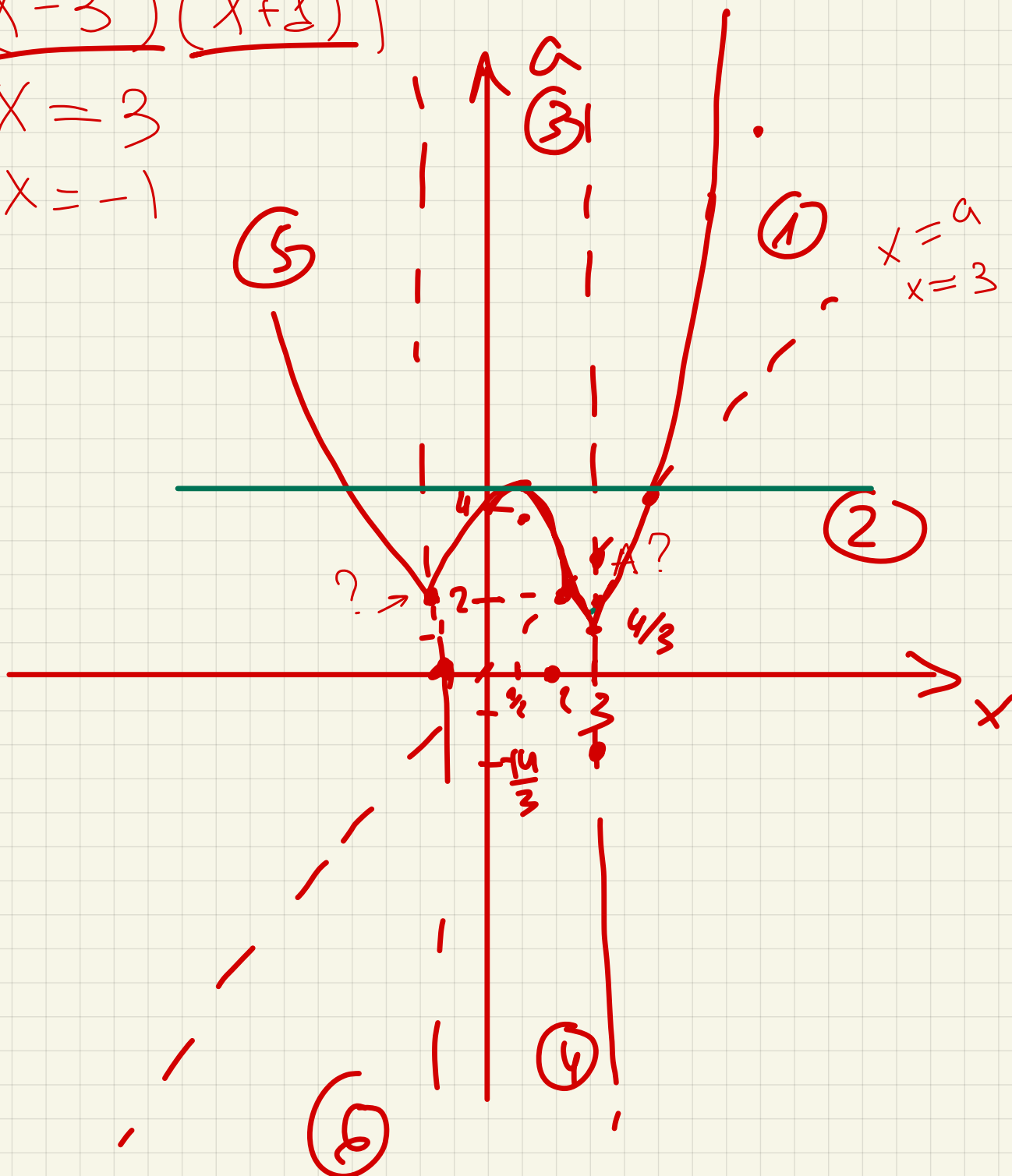
$$\checkmark x^2 - 2x - 3 \geq 0$$

$$\underline{\underline{x - a \geq 0}}$$

$$|(x-3)(x+1)|$$

$$x = 3$$

$$x = -1$$



$$\textcircled{1} \quad x^2 - 2x - 3 - 2a = x - a - 1 \quad (-3)$$

$$a = 1 + 3 - 2 = 2$$

$$x^2 - 3x - 2 = a$$

$$a = \underline{x^2 - 3x - 2}$$

$$x_0 = \frac{3}{2}$$

$$a_0 = \frac{-9}{4} - 2 = -\frac{17}{4}$$

$$A(3;3)$$

$$3 \neq -2$$

$$x^2 - 3x - 2 = x$$

$$\underline{x^2 - 4x - 2 = 0}$$

$$\frac{4 \pm \sqrt{16 + 4 \cdot 2}}{2}$$

$$x_{1,2} = \frac{4 \pm 2\sqrt{6}}{2} = 2 \pm \sqrt{6}$$

$$a = 2 \pm \sqrt{6}$$

$$\textcircled{2} \quad x^2 - 2x - 3 - 2a = -x + a - 1$$

$$x^2 - x - 2 = 3a$$

$$a = \underline{\frac{x^2 - x - 2}{3}}$$

$$x_0 = \frac{1}{2}$$

$$a_0 = \frac{\frac{1}{4} - \frac{1}{2} - 2}{3} = -\frac{3}{4}$$

$$\frac{1 \pm 3}{2} = 2$$

$$a = \frac{9-3-2}{2} = 2$$

$$3x = x^2 - x - 2$$

$$x^2 - 4x - 2 = 0$$

$$a(1-1) = \frac{1+1-2}{3} = 0$$

$$\textcircled{3} \quad -x^2 + 2x + 3 - 2a = x - a - 1$$

$$-x^2 + x + \underline{4} = a$$

$$x_0 = -\frac{1}{-2} = \frac{1}{2}$$

$$y_0 = -\frac{1}{4} + \frac{1}{2} + 4 = +\frac{1}{4} + 4 = \frac{17}{4}$$

$$(-1, 2)?$$

$$-1 - 1 + 4 = 2 \quad \textcircled{V}$$

$$x = a? \quad -x^2 + x + 4 = x$$

$$\sim x^2 + 4 = 0 \quad 1 = \pm 2$$

④

$$-x^2 + 2x + 3 - 2a = -x + a - 1$$

$$-x^2 + 3x + 4 = 3a$$

$$a = \frac{-x^2 + 3x + 4}{3}$$

$$a(2; 2)$$

$$\frac{-4 + 6 + 4}{3} = 2$$

$$a(1; 3) \Rightarrow \frac{-9 + 9 + 4}{3} = \frac{4}{3}$$

$$a \leq x - 5$$

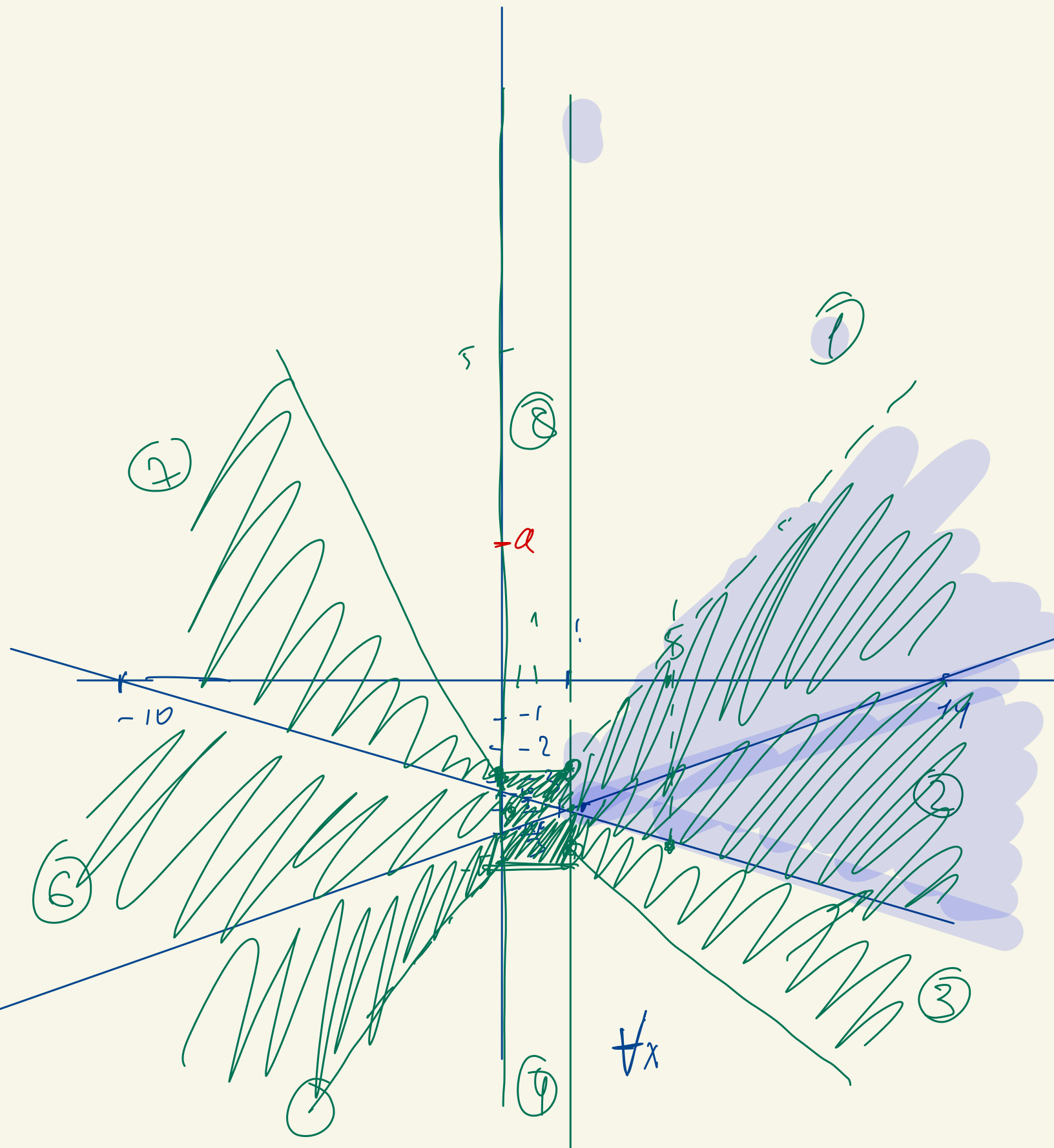
$$x + \cancel{3a} + 10 + x - \cancel{3a} - 14 \leq 6x - 6$$

$$2x - 4 \leq 6x - 6$$

$$6 - 4 \leq 4x$$

$$4x \geq 2 \\ x \geq \frac{1}{2}$$

$$|x+3a+10| + |x-3a-14| \leq 3|x| + 3|x-2|$$



$$\textcircled{1} \quad x > \frac{x-14}{3} \quad x > \frac{-x-10}{3}$$

$$x > 0 \quad x > 2$$

$$\Rightarrow x + \cancel{3a} + 10 + x - \cancel{3a} - 14 \leq 3x + 3x - 6$$

$$0 \leq x - 5$$

$$\textcircled{2} \quad a \geq \frac{-10-x}{3} \quad x - 3a - 14 \geq 0$$

$$a \leq \frac{x-14}{3}$$

$$x + \cancel{3a} + 10 + x - \cancel{3a} - 14 \leq 3x + 3x - 6$$

$$2x - 4 \leq 6x - 6$$

$$4x \geq 2$$

$$x \geq \frac{1}{2}$$

$$\textcircled{3} \quad x + 3a + 10 \leq 0$$

$$x - 3a - 14 \geq 0$$

$$a \geq -x - 3$$

$$\textcircled{4} \quad x + 3a + 10 \leq 0$$

$$x - 3a - 14 \leq 0$$

$$a \geq -5$$

$$\textcircled{5} \quad -x - 3a - 10 + x - 3a - 14 \leq -3x + 3(2-x)$$

$$x \geq x - 5$$

$$\textcircled{6} \quad -x - 3a - 10 - x + 3a + 14 \leq -3x + 3(2-x)$$

$$x \leq \frac{1}{2}$$

$$\textcircled{7} \quad 1 + 3a + 10 - x + 3a + 14 \leq -3x + 6 - 3x$$

$$a \leq -x - 3$$

$$\textcircled{8} \quad x + 3a + 10 - x + 3a + 14 \leq 3x + 3(2-x)$$

$$a \leq -3$$

$$\textcircled{9} \quad -x - 3a - 10 - x + 3a + 14 \leq 3x + 3(2-x)$$

$$x \geq -1$$

$$a \in [-5, -3]$$